

Module 1: Evolution of Computing

Slide-wise Handout of Technical Terms

Modern Computing and Python Programming (26ESCC106)

Text Book 1: Kai Hwang, Geoffrey C. Fox and Jack J. Dongarra, *Distributed and Cloud Computing*, Morgan Kaufmann, 2012 (Sections 1.1, 1.1.1, 1.1.2)

Manjunath Prasad H. R., Assistant Professor, Department of AIML, MITE

How to use this handout. Each heading matches one slide of the lecture deck (the number in the black box is the slide number shown at the bottom-right of the slide). Under each heading, the technical terms that appear on that slide are defined and explained in simple language. Diagrams are given wherever a picture makes the idea easier to understand. Terms that are explained in detail later in the course are marked (*covered later*).

Contents

Part A: Course Overview Slides (Slides 1–19)	3
Slide 1: Title Slide	3
Slide 2: Outline	3
Slide 3: Course at a Glance	3
Slide 4: Course Description	3
Slide 5: Course Objectives	4
Slide 6: Course Module 1 Topics – Modern Computing Approaches	4
Slide 7: Course Module 2 Topics – Python Fundamentals	5
Slide 8: Module 2 Laboratory – Python Basics	5
Slide 9: Course Module 3 Topics – Strings, Flow Control, and Functions	6
Slide 10: Module 3 Laboratory – Strings, Menus, and Functions	6
Slide 11: Course Module 4 Topics – Built-in Data Structures and Files	7
Slide 12: Module 4 Laboratory – Data Structures and Files	7
Slide 13: Course Module 5 Topics – Artificial Intelligence and Machine Learning	7
Slide 14: Module 5 Laboratory – Working with a Dataset	7
Slide 15: Course Outcomes	8
Slide 16: Textbooks	8
Slide 17: Reference Books and Web Links	8
Slide 18: Tools Used in this Course	8
Slide 19: Thank You	8
Part B: Module 1 – Evolution of Computing (Slides 20–49)	9
Slide 20: Module 1 – Evolution of Computing: What We Will Learn	9
Slide 21: Scalable Computing over the Internet	9
Slide 22: The Age of Internet Computing	10
Slide 23: The Platform Evolution: Five Generations	11
Slide 24: Evolutionary Trend toward Parallel, Distributed, and Cloud Computing	12
Slide 25: Reading Figure 1.1: Labels on the Arrows	13
Slide 26: High-Performance Computing (HPC)	14

Slide 27: High-Throughput Computing (HTC)	15
Slide 28: HPC versus HTC: A Comparison	16
Slide 29: Three New Computing Paradigms	16
Slide 30: How the Idea of a “Computer” Has Changed	17
Slide 31: Computing Paradigm Distinctions (1/2)	17
Slide 32: Computing Paradigm Distinctions (2/2)	18
Slide 33: Computing Paradigms at a Glance	19
Slide 34: Related Terms	19
Slide 35: Distributed System Families	20
Slide 36: Design Objectives for Future HPC and HTC Systems	21
Slide 37: Scalable Computing Trends	22
Slide 38: Degrees of Parallelism	22
Slide 39: Degrees of Parallelism: An Everyday View	24
Slide 40: Innovative Applications	24
Slide 41: Requirements of Innovative Applications	25
Slide 42: The Trend toward Utility Computing	27
Slide 43: Common Characteristics of These Paradigms	27
Slide 44: Utility Computing	28
Slide 45: The Hype Cycle of New Technologies	29
Slide 46: Five Stages of the Hype Cycle	29
Slide 47: Reading the Hype Cycle: Symbols and Examples (2010)	30
Slide 48: Summary	31
Slide 49: Review Questions	31

Part A: Course Overview Slides

Slide 1 Title Slide

No technical terms on this slide.

Slide 2 Outline

No technical terms on this slide. It lists the sections of the course overview and of Module 1.

Slide 3 Course at a Glance

Course Code

Definition: A unique alphanumeric identifier given to a course by the university.

Description: In **26ESCC106**, “26” refers to the scheme/batch year, “ES” to Engineering Science, “CC” to a common course for all branches, and “106” to its serial number.

CIE (Continuous Internal Evaluation)

Definition: Marks awarded by the institution through internal tests, assignments, and lab work during the semester.

Description: This course has 50 CIE marks.

SEE (Semester End Examination)

Definition: The final written examination conducted at the end of the semester.

Description: This course has 50 SEE marks and a 3-hour exam.

L:T:P:S

Definition: The weekly teaching pattern: **L**ecture, **T**utorial, **P**ractical (lab), and **S**elf-study hours.

Description: 3:0:2:0 means 3 hours of lecture and 2 hours of laboratory every week.

Credits

Definition: A measure of the academic weight of a course, based on contact hours.

Description: This course carries 4 credits (40 lecture hours + 24 practical hours = 64 hours).

Slide 4 Course Description

Computing Paradigm

Definition: A fundamental style or model of organizing computers to solve problems.

Description: Examples: centralized, parallel, distributed, cluster, grid, cloud, and edge computing. (Explained from Slide 31 onwards.)

Python

Definition: A high-level, general-purpose, easy-to-read programming language.

Description: Widely used in data science, AI, web development, and automation. (covered later, Module 2)

Library

Definition: A collection of ready-made code (functions, classes) that a program can reuse.

Description: Examples used in this course: NumPy, Pandas, and Scikit-learn.

Artificial Intelligence (AI)

Definition: The field of building computer systems that perform tasks normally needing human intelligence.

Description: Examples: speech recognition, chatbots, self-driving cars. *(covered later, Module 5)*

Machine Learning (ML)

Definition: A branch of AI in which computers learn patterns from data instead of being explicitly programmed.

Description: Example: an email system learning to recognize spam. *(covered later, Module 5)*

Intelligent Agent

Definition: An entity that perceives its environment through sensors and acts on it through actuators to achieve goals.

Description: Example: a robot vacuum cleaner. *(covered later, Module 5)*

Slide 5 Course Objectives

Data Type

Definition: The kind of value a variable can hold, such as integer, floating-point number, string, or Boolean.

Description: *(covered later, Module 2)*

Operator

Definition: A symbol that performs an operation on values, e.g., +, -, *, /.

Description: *(covered later, Module 2)*

Control Structure

Definition: A statement that decides the order in which instructions execute, such as if, for, and while.

Description: *(covered later, Module 3)*

File Handling

Definition: Reading data from and writing data to files stored on disk.

Description: *(covered later, Module 4)*

Slide 6 Course Module 1 Topics – Modern Computing Approaches

Cluster Computing

Definition: A group of interconnected computers working together as a single system.

Description: *(covered later in Module 1)*

Grid Computing

Definition: Sharing of computing resources across multiple organizations and locations, like an electric power grid.

Description: *(covered later in Module 1)*

Cloud Computing

Definition: Delivery of computing resources (servers, storage, software) over the Internet on demand.

Description: See Slide 32 for details.

IaaS, PaaS, SaaS

Definition: The three cloud service models: Infrastructure as a Service, Platform as a Service, and Software as a Service.

Description: Examples: virtual machines (IaaS), app-hosting platforms (PaaS), Gmail (SaaS). *(covered later in Module 1)*

Edge Computing

Definition: Processing data close to where it is generated (at the “edge” of the network) instead of in a distant data center.

Description: *(covered later in Module 1)*

Slide 7 Course Module 2 Topics – Python Fundamentals
Variable

Definition: A named location in memory that stores a value.

Keyword

Definition: A reserved word with special meaning in Python (e.g., if, for, def); it cannot be used as a variable name.

Identifier

Definition: The name given to a variable, function, or other object.

Literal

Definition: A fixed value written directly in code, e.g., 10, 3.14, "MITE".

Expression and Operator Precedence

Definition: An expression combines values and operators to produce a result; precedence decides which operator is evaluated first.

Description: Example: in $2 + 3 * 4$, multiplication is done first, giving 14.

Type Conversion

Definition: Changing a value from one data type to another, e.g., `int("25")`.

Debugging

Definition: Finding and fixing errors (bugs) in a program.

Slide 8 Module 2 Laboratory – Python Basics
USN (University Seat Number)

Definition: The unique registration number assigned to each student by the university.

Floor Division, Modulus, Exponentiation

Definition: Floor division (`//`) gives the whole-number quotient; modulus (`%`) gives the remainder; exponentiation (`**`) raises a number to a power.

Description: Example: $7 // 2 = 3$, $7 \% 2 = 1$, $2 ** 3 = 8$.

Boolean

Definition: A data type with only two values: True and False.

Slide 9 Course Module 3 Topics – Strings, Flow Control, and Functions

String, Indexing, and Slicing

Definition: A string is a sequence of characters; indexing picks one character by position; slicing picks a range of characters.

Description: Example: for `s = "Python"`, `s[0]` is 'P' and `s[0:3]` is 'Pyt'.

f-string

Definition: A formatted string literal prefixed with `f` that inserts variable values inside `{}`.

Description: Example: `f"Total = {total}"`.

Function

Definition: A named block of reusable code that performs a specific task.

Argument and Parameter

Definition: Parameters are the names listed in a function definition; arguments are the actual values passed when calling it.

Global and Local Variable

Definition: A global variable is accessible throughout the program; a local variable exists only inside the function where it is created.

Lambda Function

Definition: A small anonymous (unnamed) function written in one line, e.g., `lambda x: x*x`.

Module

Definition: A Python file containing functions and variables that can be imported into other programs.

Command Line Argument

Definition: A value passed to a program when it is started from the terminal.

Slide 10 Module 3 Laboratory – Strings, Menus, and Functions

Menu-driven Program

Definition: A program that repeatedly shows a list of options and performs the action the user selects.

Factorial

Definition: The product of all positive integers up to n : $n! = 1 \times 2 \times \dots \times n$.

Prime Number

Definition: A number greater than 1 that is divisible only by 1 and itself.

Default Argument

Definition: A parameter value that is used automatically when no argument is supplied.

Slide 11 Course Module 4 Topics – Built-in Data Structures and Files

Mutable and Immutable Objects

Definition: Mutable objects can be changed after creation (lists, sets, dictionaries); immutable objects cannot (tuples, strings, numbers).

List, Tuple, Set, Dictionary

Definition: List: ordered, changeable collection; Tuple: ordered, unchangeable collection; Set: unordered collection of unique items; Dictionary: collection of key–value pairs.

File Mode

Definition: A code that specifies how a file is opened: 'r' (read), 'w' (write), 'a' (append).

Slide 12 Module 4 Laboratory – Data Structures and Files

Set Operations

Definition: Union (all elements of both sets), intersection (common elements), difference (elements in one but not the other), symmetric difference (elements in exactly one set).

read(), readline(), readlines()

Definition: Methods that read the whole file, one line, or all lines as a list, respectively.

Slide 13 Course Module 5 Topics – Artificial Intelligence and Machine Learning

Supervised, Unsupervised, and Reinforcement Learning

Definition: Supervised: learning from labelled examples; Unsupervised: finding patterns in unlabelled data; Reinforcement: learning by trial and error using rewards.

Description: *(covered later, Module 5)*

Feature

Definition: A measurable property of the data used as input to a model (e.g., age, fare).

Data Pre-processing

Definition: Cleaning and preparing raw data (handling missing values, scaling, encoding) before training a model.

Training, Testing, and Evaluation

Definition: Training: the model learns from data; Testing: the model is checked on unseen data; Evaluation: measuring performance using metrics such as accuracy.

Slide 14 Module 5 Laboratory – Working with a Dataset

Dataset and CSV

Definition: A dataset is a structured collection of data; CSV (Comma-Separated Values) is a text file format storing tabular data.

NumPy, Pandas, Scikit-learn

Definition: NumPy: fast numerical arrays; Pandas: tables (DataFrames) for data analysis; Scikit-learn: machine learning algorithms and tools.

Target Attribute

Definition: The column the model must predict (e.g., *Survived* in the Titanic dataset).

Train–Test Split

Definition: Dividing the dataset into a training set (to learn) and a testing set (to evaluate).

Slide 15 Course Outcomes
Course Outcome (CO)

Definition: A statement of what a student should be able to do at the end of the course.

Description: Action verbs such as DESCRIBE, WRITE, IDENTIFY, and DEVELOP indicate the level of learning expected.

Slide 16 Textbooks

No technical terms on this slide.

Slide 17 Reference Books and Web Links
NPTEL

Definition: National Programme on Technology Enhanced Learning — a Government of India initiative offering free online courses from IITs and IISc.

Slide 18 Tools Used in this Course
Python 3

Definition: The current major version of the Python language, used throughout this course.

Slide 19 Thank You

No technical terms on this slide.

Part B: Module 1 – Evolution of Computing

Slide 20 Module 1 – Evolution of Computing: What We Will Learn

Scalable Computing

Definition: Computing in which a system can handle a growing amount of work simply by adding more resources (computers, processors, storage).

Description: A scalable website works equally well for 100 users and for 10 million users because more servers can be added as demand grows.

Paradigm

Definition: A model, pattern, or style of doing something.

Description: A **computing paradigm** is a particular way of organizing computers to solve problems.

HPC and HTC

Definition: High-Performance Computing and High-Throughput Computing.

Description: Explained in Slides 26–28.

Utility Computing

Definition: A model in which computing is supplied as a paid service, like electricity.

Description: Explained in Slide 44.

Hype Cycle

Definition: A graph showing how expectations about a new technology rise, fall, and settle over time.

Description: Explained in Slides 45–47.

Slide 21 Scalable Computing over the Internet

Centralized Computer

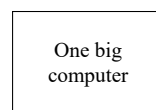
Definition: A single computer (or one tightly integrated system) that does all the processing.

Description: All work depends on one machine; if it is overloaded or fails, everything stops.

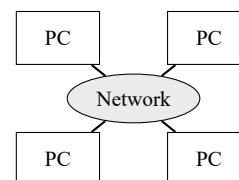
Parallel and Distributed Computing System

Definition: A system that uses multiple processors or multiple computers together to solve one large problem.

Description: The work is divided among many machines, which finish it faster and keep working even if one fails.



Centralized



Parallel / Distributed

Machine Architecture

Definition: The internal design of a computer: its processor, memory, and how they are connected.

Operating System (OS) Platform

Definition: The system software (e.g., Windows, Linux) that manages hardware and runs applications.

Network Connectivity

Definition: How computers are linked to each other to exchange data (LAN, Internet, wireless).

Application Workload

Definition: The amount and type of work that applications place on a computer system.

Data-Intensive

Definition: Describes applications whose main challenge is handling very large amounts of data rather than heavy calculation.

Description: Example: storing and searching billions of web pages.

Network-Centric

Definition: Describes computing in which the network is central; the work depends on many computers communicating.

Server

Definition: A computer that provides data or services to other computers (clients) over a network.

Slide 22 The Age of Internet Computing

Internet Computing

Definition: Computing in which services and resources are delivered over the Internet to large numbers of users.

Supercomputer

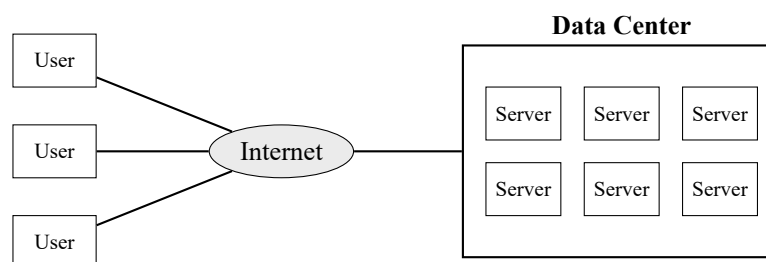
Definition: An extremely fast computer with thousands of processors, used for very complex calculations.

Description: Example: weather prediction, nuclear simulations.

Data Center

Definition: A large facility housing thousands of servers, storage systems, and networking equipment.

Description: Companies like Google, Amazon, and Microsoft run huge data centers to provide their online services.



Concurrently

Definition: At the same time.

Description: A data center serves millions of users concurrently.

Linpack Benchmark

Definition: A standard test program that measures how fast a computer can solve a large system of linear equations, reported in flops.

Description: Used to rank the world's fastest supercomputers (Top 500 list).

Benchmark

Definition: A standard test used to measure and compare the performance of computers.

Storage System

Definition: Hardware (hard disks, SSDs, storage arrays) that stores data permanently.

Bandwidth

Definition: The maximum amount of data that can be transferred over a network per second (e.g., Gbps).

Description: **High-bandwidth network** = a network that can carry a lot of data quickly.

Web Services

Definition: Software services offered over the web that other programs can use through standard protocols.

Description: Example: a payment service used by many shopping websites.

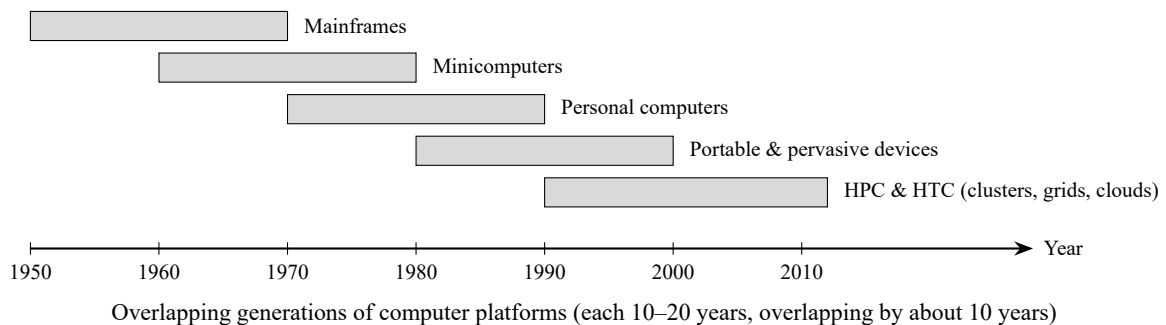
Slide 23 The Platform Evolution: Five Generations

Platform

Definition: The type of computer hardware and system software on which applications run.

Generation (of computers)

Definition: A stage in the history of computing marked by a major change in technology.



Mainframe

Definition: A large, powerful, expensive computer used by big organizations for bulk data processing.

Description: Example: IBM 360, CDC 6400.

Minicomputer

Definition: A mid-sized, lower-cost computer, smaller than a mainframe.

Description: Example: DEC PDP 11, VAX series; popular in small businesses and colleges.

Personal Computer (PC)

Definition: A computer designed for use by one person at a time.

VLSI (Very Large Scale Integration)

Definition: The technology of placing millions (now billions) of transistors on a single chip.

Description: VLSI made the microprocessor possible.

Microprocessor

Definition: A complete processor (CPU) built on a single chip.

Portable Computer and Pervasive Device

Definition: Portable computers can be carried (laptops); pervasive devices are small computing devices embedded everywhere (phones, smart cards, sensors).

Wired and Wireless

Definition: Wired: connected through cables (Ethernet); Wireless: connected through radio signals (Wi-Fi, mobile networks).

Shared Web Resources

Definition: Computing power, storage, and software on the Internet that many users share.

Slide 24 Evolutionary Trend toward Parallel, Distributed, and Cloud Computing

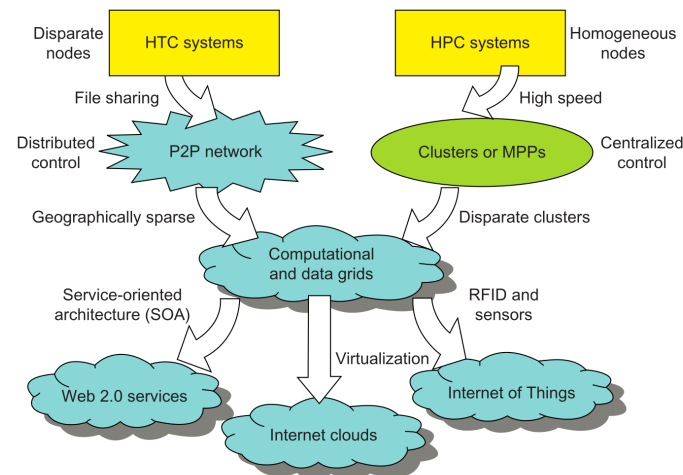


Figure 1.1 of Text Book 1

MPP (Massively Parallel Processor)

Definition: A single computer with a very large number of processors working in parallel, each with its own memory, connected by a fast internal network.

Description: Most traditional supercomputers are MPPs.

Cluster

Definition: A collection of homogeneous computers (nodes) connected closely by a fast local network and used as one system.

Description: Clusters gradually replaced MPPs because they are cheaper and allow resources to be shared.

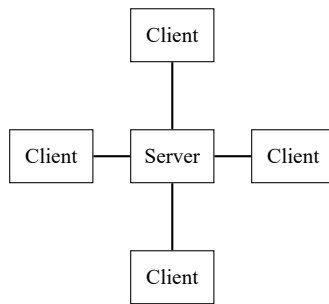
Node

Definition: A single computer in a cluster, grid, or network.

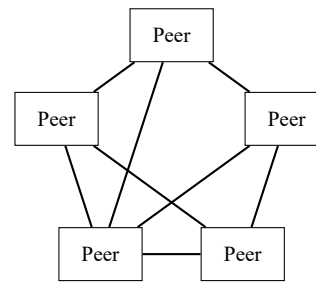
Peer-to-Peer (P2P) Network

Definition: A network in which every computer (peer) acts as both client and server, sharing resources directly without a central server.

Description: Example: BitTorrent file sharing.



Client-Server



Peer-to-Peer (P2P)

Client

Definition: A computer or program that requests services from a server.

Content Delivery

Definition: Distributing content (videos, files, web pages) to users from servers located close to them.

Computational Grid and Data Grid

Definition: A grid connects computers from many organizations and locations to share resources. A computational grid shares processing power; a data grid shares large data sets.

Web 2.0 Services

Definition: Interactive web applications where users create and share content.

Description: Examples: social networks, blogs, wikis, online maps.

Internet Cloud

Definition: A pool of computing resources in large data centers offered to users over the Internet.

Internet of Things (IoT)

Definition: A network of everyday physical objects fitted with sensors and connectivity so they can exchange data.

Description: See Slide 34.

Slide 25 Reading Figure 1.1: Labels on the Arrows

Homogeneous Nodes

Definition: Computers in a system that are all of the same type (same hardware and OS).

Description: Typical of clusters and MPPs.

Disparate (Heterogeneous) Nodes

Definition: Computers in a system that are of different types, sizes, and operating systems.

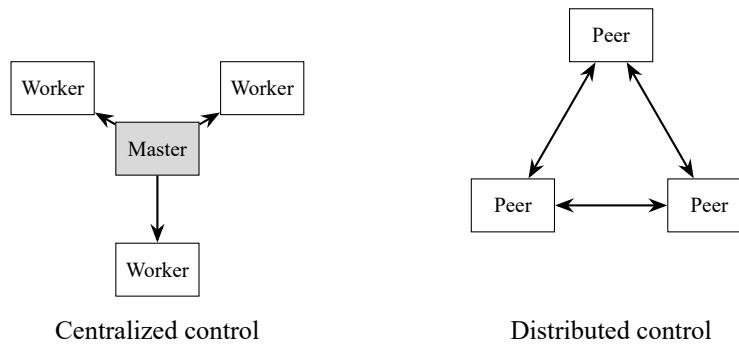
Description: Typical of P2P networks and grids.

Centralized Control

Definition: One master computer manages and coordinates all other computers.

Distributed Control

Definition: No single computer is in charge; decisions are shared among all computers.



Geographically Sparse

Definition: Spread out over a wide geographical area (different cities or countries).

Disparate Clusters

Definition: Clusters of different types owned by different organizations.

Description: When joined, they form a grid.

SOA (Service-Oriented Architecture)

Definition: A software design style in which applications are built from independent services that communicate over a network.

Description: See Slide 29.

Virtualization

Definition: Technology that creates virtual (software-based) versions of computers, storage, or networks on top of real hardware.

Description: See Slide 29.

RFID (Radio-Frequency Identification)

Definition: A technology that uses radio waves to automatically identify and track tags attached to objects.

Description: Example: FASTag on vehicles for toll payment.

Sensor

Definition: A device that detects a physical quantity (temperature, motion, light) and converts it into data.

Slide 26 High-Performance Computing (HPC)

High-Performance Computing (HPC)

Definition: The use of supercomputers and clusters to solve large, complex problems as **fast** as possible.

Description: The focus is on raw speed for a single big problem.

Raw Speed Performance

Definition: The pure calculating speed of a computer, without considering cost or number of users.

Flops (Floating-Point Operations per Second)

Definition: The number of calculations with decimal numbers a computer performs every second.

Description: The standard unit of supercomputer speed.

Unit	Prefix	Value	Era (from textbook)
Gflops	Giga	10^9 flops	Early 1990s
Tflops	Tera	10^{12} flops	Around 2000
Pflops	Peta	10^{15} flops	2010

Floating-Point Number

Definition: A number with a decimal point, such as 3.14 or 6.02×10^{23} .

Top 500

Definition: A list, updated twice a year, of the 500 most powerful computers in the world, ranked by Linpack benchmark results.

Simulation

Definition: Using a computer model to imitate a real-world process (weather, earthquakes) to study it.

Genome Analysis

Definition: Computer processing of DNA sequence data to study genes.

Slide 27 High-Throughput Computing (HTC)

High-Throughput Computing (HTC)

Definition: Computing that aims to complete the **largest number of tasks** in a given time, rather than one task as fast as possible.

Description: Suited to serving millions of independent requests, such as web searches.

Throughput

Definition: The number of tasks (jobs, requests) completed per unit of time.

Description: Example: 10,000 search requests answered per second.

High-Flux Computing

Definition: Another name for HTC: computing with a very high flow (flux) of tasks passing through the system.

Batch Processing

Definition: Running a group (batch) of jobs together without user interaction.

Description: Example: generating salary slips of all employees at month end.

Reliability

Definition: The ability of a system to keep working correctly without failure over time.

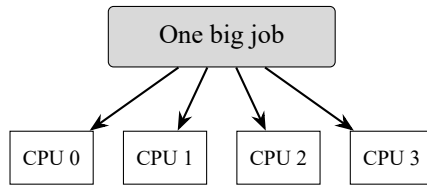
Security

Definition: Protection of systems and data from unauthorized access and attacks.

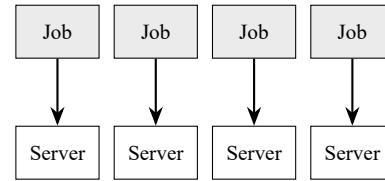
Energy Saving

Definition: Reducing the electricity consumed by servers and data centers.

Slide 28 HPC versus HTC: A Comparison



HPC: one job split across many processors → finish fast



HTC: many independent jobs → more jobs per hour

Job / Task

Definition: A unit of work submitted to a computer system.

Analogy: Racing Car vs Metro Train

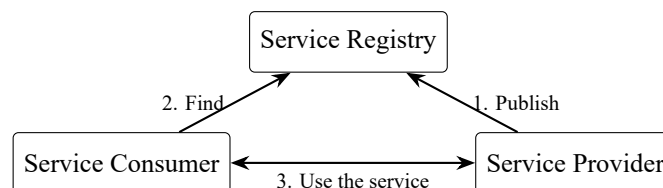
Definition: HPC is like a racing car: it carries one passenger very fast. HTC is like a metro train: each trip is not the fastest, but it carries thousands of passengers per hour.

Slide 29 Three New Computing Paradigms

Service-Oriented Architecture (SOA)

Definition: A way of building software as a collection of independent **services**, each doing one job, that communicate over a network using standard interfaces.

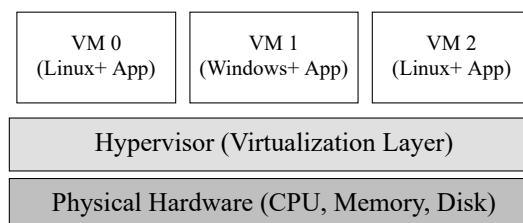
Description: A service provider publishes its service in a registry; a service consumer finds it and uses it.



Virtualization

Definition: Creating several virtual machines (VMs) on one physical computer using software called a **hypervisor**; each VM behaves like a separate computer with its own OS.

Description: Virtualization is the key technology behind cloud computing.



Virtual Machine (VM)

Definition: A software-based computer that runs on a physical machine and behaves like a real computer.

GPS (Global Positioning System)

Definition: A satellite-based system that gives the exact location of a receiver anywhere on Earth.

Paradigms Blurring

Definition: The boundaries between clusters, grids, P2P systems, and clouds are becoming less clear, since clouds are often built from clusters with virtualization added.

Slide 30 How the Idea of a “Computer” Has Changed

Computer Utility

Definition: The idea that computing can be supplied to homes and offices like a public utility (electricity, water, telephone).

Description: Predicted by Leonard Kleinrock in 1969; realized today by cloud computing.

“The Network is the Computer”

Definition: John Gage’s (Sun Microsystems, 1984) idea that the real power lies in many networked computers, not in one machine.

“The Data Center is the Computer”

Definition: David Patterson’s (2008) idea that an entire data center should be designed and programmed as one giant computer.

“The Cloud is the Computer”

Definition: Rajkumar Buyya’s idea that users simply use the cloud, without caring which physical machine does the work.

Software as a Service

Definition: Software that runs on the provider’s servers and is used through the Internet instead of being installed on the user’s PC.

Description: Example: Google Docs versus installing a word processor.

Slide 31 Computing Paradigm Distinctions (1/2)

Centralized Computing

Definition: A paradigm in which all computer resources — processors, memory, and storage — are in one physical system, fully shared and controlled by one operating system.

Tightly Coupled

Definition: Components that are closely connected and depend heavily on each other, typically sharing memory.

Loosely Coupled

Definition: Components that are connected but work fairly independently, each with its own memory.

Integrated OS

Definition: A single operating system that manages all resources of the system.

Parallel Computing (Parallel Processing)

Definition: Using many processors at the same time to solve one problem faster by dividing it into parts.

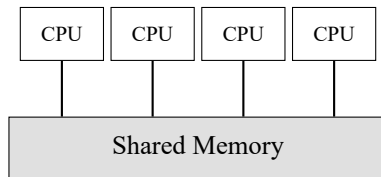
Shared Memory

Definition: One memory that all processors can read and write directly.

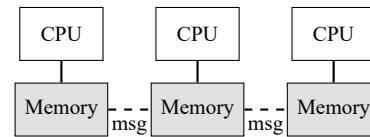
Description: Processors communicate by writing to and reading from common memory locations.

Distributed Memory

Definition: Each processor has its own private memory; processors communicate by sending messages.



Shared memory (tightly coupled)



Distributed memory (loosely coupled)

Interprocessor Communication

Definition: The exchange of data between processors working on the same problem.

Message Passing

Definition: A method of communication in which processes exchange data by explicitly sending and receiving messages.

Parallel Computer, Parallel Program, Parallel Programming

Definition: A parallel computer can run many computations at once; a parallel program is software written for it; parallel programming is the process of writing such programs.

Slide 32 Computing Paradigm Distinctions (2/2)

Distributed Computing

Definition: The field of computer science that studies distributed systems.

Distributed System

Definition: A system of multiple **autonomous** computers, each with its own private memory, that communicate through a computer network and work together.

Description: To the user, the whole system appears as one.

Autonomous Computer

Definition: A computer that is independent: it has its own processor, memory, and OS and can work on its own.

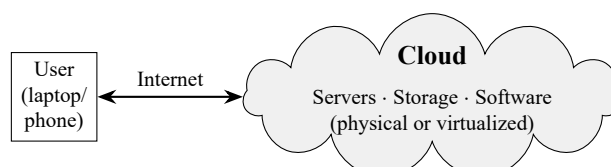
Distributed Program and Distributed Programming

Definition: A distributed program runs across many computers in a distributed system; writing such programs is distributed programming.

Cloud Computing

Definition: A paradigm in which a pool of physical or virtualized resources in large data centers is offered to users over the Internet, using parallel and/or distributed computing.

Description: Users pay only for what they use and need not own the hardware.



Physical vs Virtualized Resources

Definition: Physical resources are real hardware; virtualized resources are software-created versions (VMs) running on that hardware.

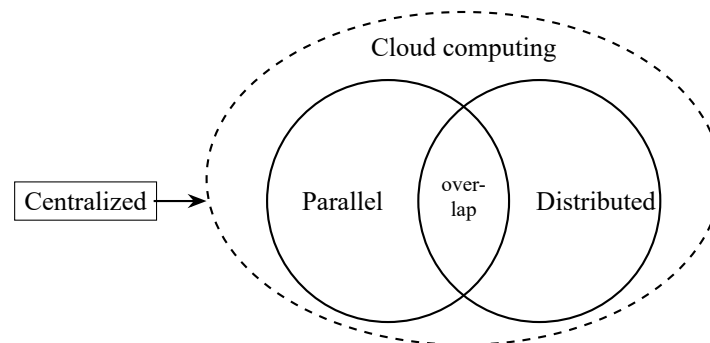
Service Computing

Definition: Computing in which everything is offered and used as a service over the network.

Slide 33 Computing Paradigms at a Glance

Overlap of Paradigms

Definition: The paradigms are not completely separate: parallel computing overlaps with distributed computing, and cloud computing uses ideas from centralized, parallel, and distributed computing.



Distributed computing is the opposite of centralized computing; the paradigms overlap.

Opposite of Centralized

Definition: Distributed computing spreads resources over many computers, whereas centralized computing keeps them in one system.

Slide 34 Related Terms

Concurrent Computing / Concurrent Programming

Definition: Executing several computations during overlapping time periods; usually refers to the union of parallel and distributed computing.

Ubiquitous Computing

Definition: Computing that is available everywhere and at any time, using small pervasive devices with wired or wireless communication.

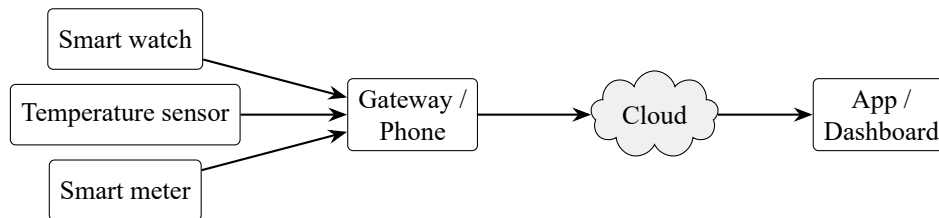
Description: Example: smartphones, smart watches, smart home devices.

Pervasive Device

Definition: A small computing device embedded in everyday surroundings.

Internet of Things (IoT)

Definition: A networked connection of everyday objects — computers, sensors, machines, even humans — that collect and exchange data, supported by Internet clouds.



Internet Computing

Definition: The broadest term, covering all computing paradigms over the Internet.

Slide 35 Distributed System Families

Distributed System Family

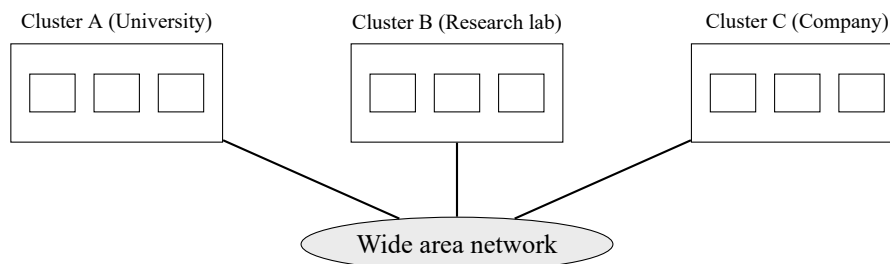
Definition: A group of related distributed system types: clusters, grids, P2P networks, and clouds.

Wide Area Computing Infrastructure

Definition: Computing facilities connected across a large geographical area (a country or the world).

Grid (Computational / Data Grid)

Definition: A system that connects clusters and computers owned by different organizations across many locations so they can share computing power and data.



A grid joins clusters owned by different organizations over a wide area network.

Service-Oriented Computing

Definition: Computing in which software functions are provided as services over a network.

Resource Sharing

Definition: Allowing many users or organizations to use the same hardware, software, and data.

Massively Distributed System

Definition: A distributed system made of a very large number (thousands to millions) of machines.

Parallelism / Concurrency

Definition: Parallelism: doing many things at exactly the same time on different processors. Concurrency: managing many tasks in overlapping time.

CPU Core

Definition: An individual processing unit inside a CPU; a multicore CPU has several cores.

GPU (Graphics Processing Unit)

Definition: A processor with thousands of small cores, originally for graphics, now used for fast parallel calculations.

Description: Tianhe-1A had over 3.2 million GPU cores.

Tianhe-1A

Definition: A Chinese supercomputer that was the fastest cluster machine in the world in October 2010.

Slide 36 Design Objectives for Future HPC and HTC Systems

Multicore / Many-core Processor

Definition: A processor chip with several (multicore) or very many (many-core) processing cores.

Thread

Definition: The smallest sequence of instructions that can be scheduled and run independently by a processor.

Description: A program can have many threads running at once.

Throughput per Watt

Definition: The amount of work done for each unit of electrical power consumed; a measure of energy efficiency.

Efficiency

Definition: How well a system uses its resources (processors, memory, power) to get work done.

Utilization Rate

Definition: The fraction of time a resource is actually busy doing useful work.

Dependability

Definition: The ability of a system to deliver service that can be trusted — reliable, available, and self-managing — even when parts fail.

Self-Management

Definition: The ability of a system to monitor, configure, and repair itself automatically.

Quality of Service (QoS)

Definition: A guaranteed level of performance, such as response time, availability, and throughput.

Programming Model

Definition: The way a programmer is expected to write programs for a system (e.g., parallel or distributed programming model).

Adaptation

Definition: The ability of the programming model to handle huge numbers of requests and data sets on virtualized cloud resources.

Flexibility in Application Deployment

Definition: The ability of a system to run both scientific (HPC) and business (HTC) applications well.

Deployment

Definition: Installing and running an application on a system so that users can use it.

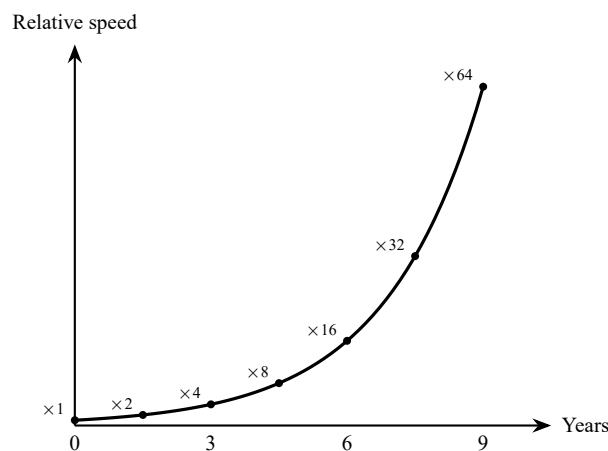
Slide 37 Scalable Computing Trends

Rules of Thumb in Data Engineering

Definition: A well-known paper by Jim Gray describing practical trends in storage, processing, and networking costs.

Moore's Law

Definition: The observation that processor speed (originally, the number of transistors on a chip) doubles roughly every 18 months.



Moore's law: doubling every 1.5 years gives 64 times the speed in 9 years.

Gilder's Law

Definition: The observation that network bandwidth has doubled every year.

Commodity Hardware

Definition: Ordinary, mass-produced, low-cost computer components (like those in desktops and laptops).

Description: Large clusters and data centers are built from commodity hardware because it is cheap.

Price/Performance Ratio

Definition: The cost of a system divided by its performance; a lower value means better value for money.

Degree of Parallelism (DoP)

Definition: The number of operations or tasks that can be executed at the same time.

Resource Distribution

Definition: Spreading computing resources across many machines and locations.

Slide 38 Degrees of Parallelism

Bit-Serial Processing

Definition: Processing data one bit at a time.

Bit-Level Parallelism (BLP)

Definition: Processing many bits at once by increasing the word size of the processor (4 → 8 → 16 → 32 → 64 bits).

Word

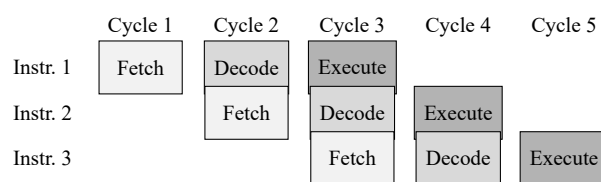
Definition: The number of bits a processor handles as one unit (e.g., 64 bits on modern CPUs).

Instruction-Level Parallelism (ILP)

Definition: Executing several machine instructions of a program at the same time inside one processor.

Pipelining

Definition: Splitting instruction execution into stages (fetch, decode, execute) so that different instructions are in different stages at the same time, like an assembly line.



Pipelining: in cycle 3, three instructions are being processed at once.

Superscalar Processor

Definition: A processor that can issue and execute more than one instruction per clock cycle using multiple execution units.

VLIW (Very Long Instruction Word)

Definition: An architecture in which the compiler packs several independent operations into one long instruction that executes in parallel.

Multithreading

Definition: Running multiple threads on a processor so that its resources are kept busy.

Branch Prediction, Dynamic Scheduling, Speculation

Definition: Techniques that help ILP: guessing the outcome of if decisions in advance, reordering instructions at run time, and executing instructions before it is certain they are needed.

Compiler

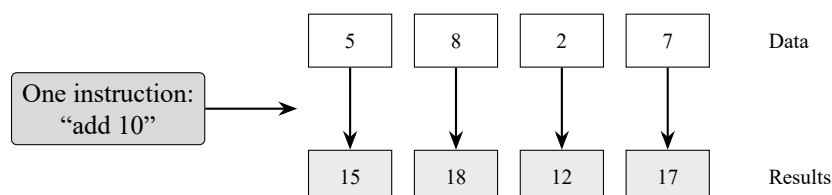
Definition: A program that translates source code written by a programmer into machine instructions.

Data-Level Parallelism (DLP)

Definition: Applying the same operation to many data items at the same time.

SIMD (Single Instruction, Multiple Data)

Definition: A processor design in which one instruction operates on many data elements simultaneously.



SIMD: one instruction processes four data items at the same time.

Vector Machine

Definition: A processor whose instructions work on whole arrays (vectors) of numbers at once.

Task-Level Parallelism (TLP)

Definition: Running different tasks (parts of a program) on different cores at the same time.

Chip Multiprocessor (CMP)

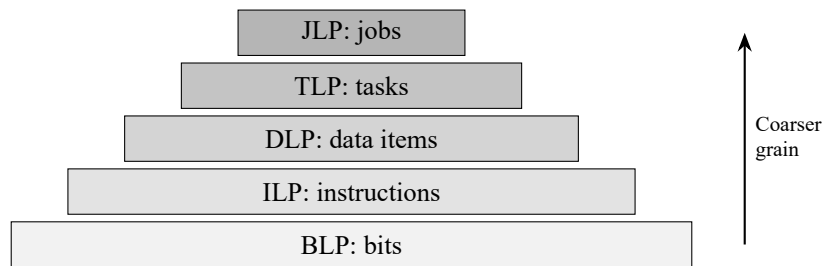
Definition: A single chip containing multiple processor cores (a multicore processor).

Job-Level Parallelism (JLP)

Definition: Running entire independent jobs on different computers at the same time in a distributed system.

Granularity (Fine-grain vs Coarse-grain)

Definition: The size of the work units executed in parallel. Fine-grain: small units such as bits and instructions; coarse-grain: large units such as tasks and jobs.



Coarse-grain parallelism is built on top of fine-grain parallelism.

Slide 39 Degrees of Parallelism: An Everyday View

Pixel

Definition: The smallest dot of colour in a digital image.

Description: Brightening every pixel of a photo with one instruction is an example of DLP.

Assembly Line

Definition: A production method in which a product moves through stations, each doing one step; used as an analogy for pipelining (ILP).

Parallel Processing vs Distributed Processing

Definition: Parallel processing happens inside one computer (many processors/cores); distributed processing happens across many computers on a network.

Description: Granularity increases from parallel to distributed processing.

Slide 40 Innovative Applications

Telecommunication

Definition: Communication over a distance using electronic means (telephone, mobile, Internet).

E-commerce

Definition: Buying and selling goods and services over the Internet.

Transaction Processing

Definition: Handling business operations (transactions) such as payments and bookings so that each is completed fully and correctly.

Telemedicine

Definition: Providing medical consultation and care remotely using telecommunication technology.

Distance Education

Definition: Teaching students who are not physically present, using online platforms.

Electric Power Grid

Definition: The network that delivers electricity from power plants to consumers; it is monitored and controlled by computers.

Decision-Making System

Definition: A computer system that analyzes data to help people make decisions.

Worm Containment

Definition: Detecting and stopping the spread of computer worms (self-copying malicious programs) across networks.

Cyber Security

Definition: Protecting computers, networks, and data from digital attacks.

Digital Government (e-Governance)

Definition: Delivering government services online.

Description: Example: online income tax return filing.

Mission-Critical Application

Definition: An application whose failure can cause serious loss of life, money, or security.

Description: Examples: military command and control, crisis management.

Intelligent System

Definition: A computer system that can learn, reason, or adapt using AI techniques.

Slide 41 Requirements of Innovative Applications
Transparency

Definition: Hiding the complexity of a distributed system so that users see it as one simple system.

Type of Transparency The user does not need to know...

Data access	where or how the data is stored
Resource allocation	which machines are given to the job
Process location	on which computer the program is running
Concurrency	that many other users are running jobs at the same time
Job replication	that copies of the job are running for safety
Failure recovery	that a machine failed and the system recovered

Resource Allocation

Definition: Assigning processors, memory, and storage to jobs.

Replication

Definition: Keeping copies of data or jobs on multiple machines for reliability and speed.

Failure Recovery

Definition: Restoring normal operation automatically after a hardware or software failure.

Computing Economics

Definition: Achieving the required computing at the lowest possible cost.

Web-Scale Data Collection

Definition: Collecting data from the entire web or from millions of users.

Scalable Performance

Definition: Performance that grows in proportion as more resources are added.

Distributed Transaction

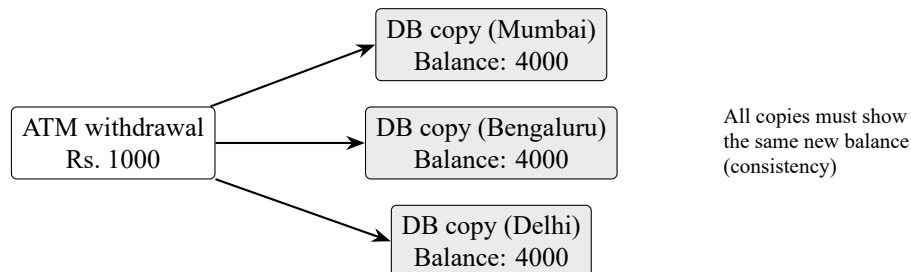
Definition: A transaction that updates data stored on two or more database servers; it must succeed on all of them or on none.

Database Server

Definition: A server that stores and manages a database and answers queries.

Consistency of Replicated Records

Definition: Ensuring that all copies of the same data on different servers have the same value at all times.



Network Saturation

Definition: A condition in which a network is so full of traffic that it cannot carry more data, causing delays.

Security Threat

Definition: Any possible danger that can exploit a weakness to harm a system (hacking, viruses, data theft).

Slide 42 The Trend toward Utility Computing

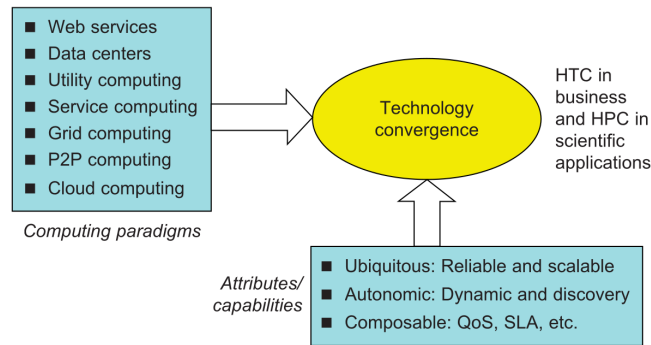


Figure 1.2 of Text Book 1

Service Computing

Definition: Computing in which applications are composed from services offered over the network.

P2P Computing

Definition: Computing performed by peers that share resources directly with each other.

Technology Convergence

Definition: Different technologies coming together and merging into a common platform.

Description: All the computing paradigms in Figure 1.2 converge to support HTC in business and HPC in science.

Slide 43 Common Characteristics of These Paradigms

Ubiquitous

Definition: Present everywhere in daily life.

Description: Requires the systems to be **reliable** and **scalable**.

Scalability

Definition: The ability of a system to handle more work by adding more resources without a drop in performance.

Autonomic

Definition: Able to manage itself automatically — configuring, healing, and organizing — without human intervention.

Self-Organization

Definition: The ability of a system to arrange and reorganize its components automatically.

Dynamic Discovery

Definition: Automatically finding available resources or services at run time.

Composable

Definition: Able to be combined with other services to build larger applications.

SLA (Service-Level Agreement)

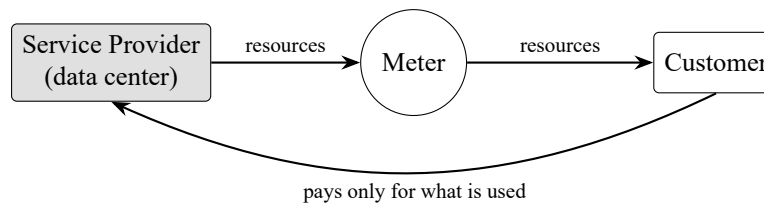
Definition: A formal contract between a service provider and a customer that specifies the level of service guaranteed (e.g., 99.9% availability) and penalties if not met.

Slide 44 Utility Computing

Utility Computing

Definition: A business model in which customers obtain computing resources from a paid service provider and pay according to usage.

Description: Like an electricity meter: you pay for the units you consume.



Service Provider

Definition: A company that offers computing resources or services to customers.

Description: Examples: Amazon Web Services, Microsoft Azure, Google Cloud.

Edge Network

Definition: The part of a network close to the end users and devices.

Distributed OS

Definition: An operating system that manages a group of networked computers and makes them appear as one system.

Middleware

Definition: Software that sits between the operating system and applications, connecting distributed components.

Resource Management

Definition: Allocating, scheduling, and monitoring computing resources efficiently.

Slide 45 The Hype Cycle of New Technologies

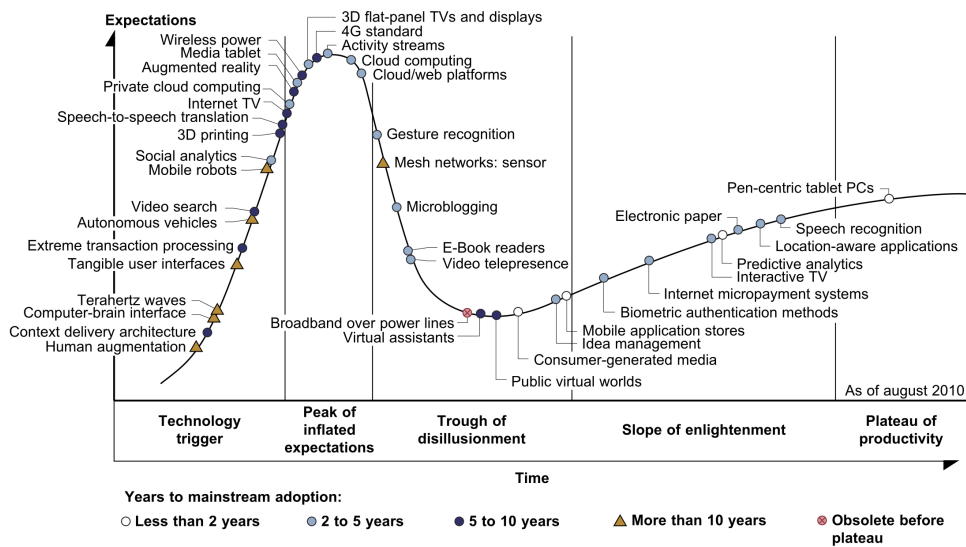


Figure 1.3 of Text Book 1 (source: Gartner, Inc., 2010)

Hype

Definition: Exaggerated publicity and excitement about something.

Hype Cycle

Definition: A graphical representation (by the research firm Gartner) of how expectations about an emerging technology change over time until it is widely adopted.

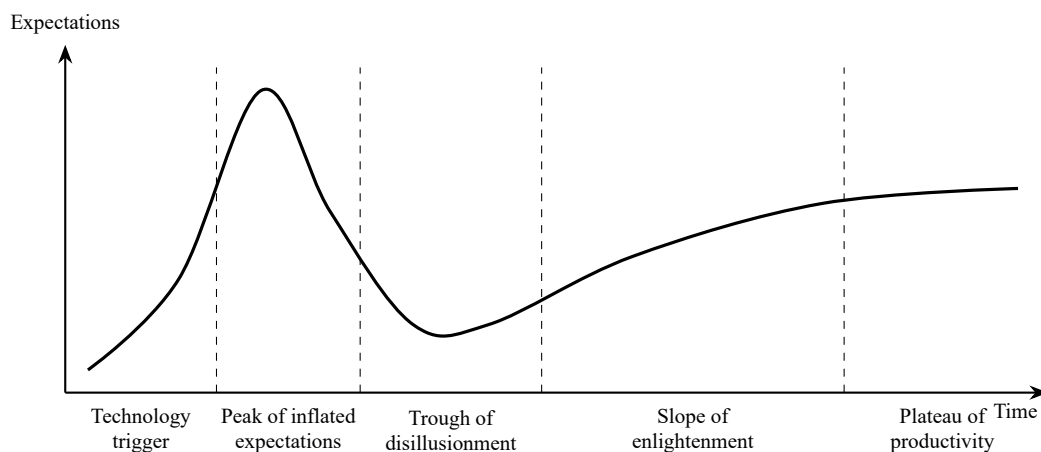
Emerging Technology

Definition: A new technology that is still developing and not yet widely used.

Mainstream Adoption

Definition: The stage when a technology is used widely by the general public and businesses.

Slide 46 Five Stages of the Hype Cycle



Simplified hype cycle showing its five stages.

Technology Trigger

Definition: The stage when a new technology is first introduced or demonstrated and starts attracting attention.

Peak of Inflated Expectations

Definition: The stage when publicity is at its highest and expectations are often unrealistically high.

Trough of Disillusionment

Definition: The stage when the technology fails to meet the inflated expectations and interest drops.

Slope of Enlightenment

Definition: The stage when people better understand the real benefits and practical uses, and expectations rise steadily.

Plateau of Productivity

Definition: The stage when the technology becomes stable, widely accepted, and productively used.

Slide 47 Reading the Hype Cycle: Symbols and Examples (2010)
Years to Mainstream Adoption

Definition: The expected time a technology needs to reach the plateau of productivity, shown by the symbol on the curve.

Description: Hollow circle: less than 2 years; gray circle: 2–5 years; dark circle: 5–10 years; triangle: more than 10 years; crossed circle: obsolete before reaching the plateau.

Obsolete

Definition: No longer used because it has been replaced by something better.

Description: Example: broadband over power lines (2010).

Consumer-Generated Media

Definition: Content such as videos, reviews, and blogs created by ordinary users rather than professionals.

Internet Micropayment

Definition: A system for making very small online payments (e.g., paying a few rupees to read an article).

3D Printing

Definition: Making a physical object layer by layer from a digital 3D model.

Mesh Network Sensors

Definition: Sensors that connect to each other directly in a mesh, passing data along from node to node.

Broadband over Power Lines

Definition: Delivering high-speed Internet through ordinary electrical power cables.

Biometric Authentication

Definition: Verifying a person's identity using physical features such as fingerprints, face, or iris.

Description: Example: Aadhaar-based fingerprint authentication.

Speech Recognition

Definition: Technology that converts spoken words into text or commands.

Predictive Analytics

Definition: Using data and statistical models to predict future events.

Media Tablet

Definition: A portable touch-screen computer such as an iPad.

Interactive TV

Definition: Television that allows viewers to interact with content (vote, shop, choose programs).

Slide 48 Summary

This slide revisits terms already defined. Quick recap:

Term	One-line meaning	Slide
HPC	Solve one big problem as fast as possible	26
HTC	Complete as many tasks as possible per unit time	27
SOA / Virtualization / IoT	Enablers of Web 2.0, clouds, and connected objects	29
Centralized computing	All resources in one system	31
Parallel computing	Many processors on one problem	31
Distributed computing	Many autonomous computers with private memory	32
Cloud computing	Resources over the Internet, centralized or distributed	32
BLP → JLP	Degrees of parallelism from bits to jobs	38
Ubiquitous, Autonomic, Composable	Common attributes of modern paradigms	43
Utility computing	Pay-per-use computing from a provider	44
Hype cycle	Five-stage curve of technology expectations	46

Slide 49 Review Questions

No new terms. Pointers to the slides that answer each question:

Q.	Question	Refer to
1	Five generations of computer platforms with examples	Slide 23
2	Differentiate HPC and HTC	Slides 26–28
3	Evolutionary trend in Figure 1.1	Slides 24–25
4	Centralized, parallel, distributed, and cloud computing	Slides 31–33
5	Three new computing paradigms and their enablers	Slide 29
6	Five degrees of parallelism	Slides 38–39
7	Vision of computer utilities (Figure 1.2)	Slides 42–44
8	Five stages of the hype cycle	Slides 45–47